



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS

Final Year Project Report on
"DESIGN, FABRICATION, AND TESTING OF AN ADJUSTABLE FRUITS
SORTING MACHINE"

Prepared by	
Abhijit Shrestha	THA078BIE003
Bimal Gurung	THA078BIE008
Birendra Prasad Shah	THA078BIE017
Sabin Ghimire	THA078BIE032

A FINAL YEAR PROJECT REPORT
SUBMITTED TO THE DEPARTMENT OF INDUSTRIAL ENGINEERING
IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
DEGREE OF BACHELOR IN
INDUSTRIAL ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING
KATHMANDU, NEPAL

December, 2025

Copyright

The author has agreed that the library, Department of Industrial Engineering Thapathali Campus, Institute of Engineering may make this report freely available for inspection. Moreover, the author has agreed that permission for extensive copying of this report for scholarly purpose may be granted by the professor(s) who supervised the work recorded herein or, in their absence, by the Head of the Department wherein the report was done. It is understood that the recognition will be given to the author of this report and to the Department of Industrial Engineering, Thapathali Campus, and Institute of Engineering in any use of the material of this report. Copying or publication or the other use of this report for financial gain without approval of the Department of Industrial Engineering, Thapathali Campus, Institute of Engineering and author's written permission is prohibited. Request for permission to copy or to make any other use of the material in this report in whole or in part should be addressed to:

Head of Department

Department of Industrial Engineering

Thapathali Campus, Institute of Engineering

Thapathali, Kathmandu, Nepal

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS

It is hereby certified that this paper, entitled
“A Project report on "**DESIGN, FABRICATION, AND TESTING OF AN**
ADJUSTABLE FRUITS SORTING MACHINE"

Prepared by
Abhijit Shrestha, Bimal Gurung, Birendra Prasad Shah, and Sabin Ghimire
is an outcome of the research and fabrication conducted under
“Department of Industrial Engineering”

The facts and ideas presented in this paper are an outcome of student’s hard work and
dedication to the project, undertaken as a partial fulfillment for requirements for degree of
Bachelors in Industrial Engineering.

The outcome of this project has been highly appreciated.

Head of Department
Er. Kamal Pokhrel
Department of Industrial Engineering

Project Supervisor
Er. Suman Acharya

External Evaluator
Er. Reewaj Bhandari

Date: December, 2025

Abstract

This research aims to develop and fabricate an adjustable fruit sorting machine which simple, does not require an electric power to operate and has low cost of fabrication. The design and fabrication of project was completed at workshop of Thapathali Campus. After fabrication of machine, the testing of machine was done by taking sample of 5kg oranges, 6 kg apples, and 6 kg guavas. On average, the machine sorted oranges 6.28 times, apples 8.14 times, and guavas 8.27 times faster than manual methods. Similarly, the gap between pipes were divided into three steps i.e. maximum gap, medium gap, and minimum gap and corresponding range of fruits that can be sorted in three compartments i.e. large, medium, and small was found to be 14-92 mm, 14-75 mm and 14-49mm respectively.. The calculation of angle calibration shows when moving pipe by 1mm the angle changes by 0.034 degree which establish the relation between linear movements of pipe with angle change while adjusting the gap between pipes. The inclination of pipe can with the vertical is adjustable from 85^0 to 81^0 which is very crucial to control flow rate of fruits. The machine was fabricated using locally available materials like mild steel angles square rods, flat bars, metal sheet, etc., by using machine like drilling machine, grinding, machine, welding machine, riveting tools, measuring tools, etc. The machine was able sort different fruits like oranges, apples, and guavas etc. which were of different size which shows the design and fabrication of machine was successful. There are some limitations of machine like manual feeding, small size, absence of cushioning and counting sensor, etc. However, the machine was working efficiently and there is scope of future improvements which will make it affordable, simple, user friendly machine which runs without any electric power and can be commercialized in Nepali agriculture markets.

Keywords: sorting, gap, step, machine, adjustable, hopper, efficiency, etc.

Acknowledgement

We want to express our sincere gratitude towards **Er. Kamal Pokhrel**, the head of the Department of Industrial Engineering, Thapathali Campus, for his support and coordination. We are deeply indebted to our project supervisor, **Er. Suman Acharya** for his support, guidance and mentorship. His valuable suggestions and unwavering support have made the project feasible. We got insightful ideas, suggestions, tips, and expertise from him, which played a key role when we were in a dilemma and confusion. His dedication, patience and expert advice motivated us to improve and complete this project. We truly appreciate the time and effort he put into helping us complete this project successfully.

We are deeply indebted to **Er. Ankit Shrestha, Er. Pitambar Chaulagain, Er. Sudan Naupane, Er Bishworam Parajuli, and Er. Rajan Sharma** of the Department of Industrial Engineering, Thapathali Campus, for their constant guidance, mentoring, insightful knowledge, and constructive criticism.

At last, we want to thank all the writers whose work has been cited in this report and our friends and seniors for their support, help, and encouragement.

Project Members

Abhijit Shrestha

Bimal Gurung

Birendra Prasad shah

Sabin Ghimire

Table of Contents

Copyright.....	i
Abstract	iii
Acknowledgement	iv
Table of Contents	v
List of tables.....	vii
List of figures	viii
List of acronyms and abbreviations	ix
CHAPTER ONE:	1
INTRODUCTION	1
1.1 Background of Project.....	1
1.2 Statement of Problem	3
1.3 Objectives.....	4
1.3.1 General Objective	4
1.3.2 Specific Objectives	4
1.4 Scope of Sorting Machine	4
1.5 Key Features.....	5
CHAPTER TWO:	6
LITERATURE REVIEW	6
CHAPTER THREE:	9
RESEARCH METHODOLOGY.....	9
3.1 Theoretical Framework	9
3.2 Data Collection.....	9
3.2.1 Primary data Collection	9
3.2.2 Secondary Data Collection.....	9
3.3 Study Design	10
CHAPTER FOUR:.....	12
DESIGN AND FABRICATION	12
4.1 Preliminary Design.....	12
4.2 Design in SolidWorks	12
4.3 Components Design and Fabrication	12

4.4	Assembly of each components	16
4.5	Machine Used in Fabrication	17
CHAPTER FIVE:		18
CALIBRATION AND TESTING		18
5.1	Calibration of Machine.....	18
5.2	Angle calibration	19
5.3	Inclination (Slope) adjustment mechanism of pipes	21
5.4	Testing of machine	21
5.4.1	Manual sorting	21
5.4.2	Sorting by machine	23
5.5	Comparison between Manual and Machine Sorting	24
CHAPTER SIX:		25
BUDGET OF PROJECT		25
CHAPTER SEVEN:		26
RESULT AND DSSUSSION		26
7.1	Result Presentation	26
7.2	Analysis of Result	26
7.3	Outcomes	27
7.4	Precautionary Measures	27
7.5	Limitations of Project.....	28
7.5	Future improvement	28
CHAPTER EIGHT:		29
CONCLUSION AND RECOMMENDATION		29
8.1	Conclusion.....	29
8.2	Recommendations	29
Bibliography.....		30
ANNEXES		32

List of tables

Table 1.1: Statistics of orange production in economic year 2076/77.....	2
Table 5.1: Calibration table for adjusting the pipes.....	18
Table 5.2: Angle calibration table per mm.....	20
Table 5.3.1: Manual sorting time of different fruits by different people	22
Table 5.3.2: Sorting time taken by machine for different fruits at different angle.....	23
Table 5.5(a): Comparison of manual sorting and machine sorting for 5 kg oranges.....	24
Table 5.5(b): Comparison of manual sorting and machine sorting for 6 kg apples.....	24
Table 5.5(c): Comparison of manual sorting and machine sorting for 6 kg guavas.....	24
Table 6: Budget of the project.....	25

List of figures

Figure 1.1(a): Fruit grading machine with price	1
Figure 1.1(b): Fruit grading machine with price.....	1
Figure 1.1(c): Fruit grading machine with price	1
Figure 4.3(a): Design of frame before	12
Figure 4.3(b): Design of frame after	12
Figure 4.3(c): Actual frame fabrictaed in workshop.....	12
Figure 4.3(d): Design of hopper before	13
Figure 4.3(e): Design of hopper after	13
Figure 4.3(f): Design of hopper sheet	13
Figure 4.3(g): Actual fabrication hopper frame in workshop	13
Figure 4.3(h): Design of sliding sheet.....	14
Figure 4.3(i): Detailed design of sliding sheet.....	14
Figure 4.3(j): Design of slider for pipes.....	14
Figure 4.3(k): Deatiel design of slider for pipes	15
Figure 4.3(l): Aluminium pipes used for sorting	15
Figure 4.3(m): M5 nut,bolt and washer	15
Figure 4.3(n): Square rod used to make frame stable	15
Figure 4.3(o): Assembly of all components in SolidWorks	16
Figure 4.3(p): Actual machine fabrictaed in workshop	16
Figure 4.3(q): Rotating view of assembly in SolidWorks	16
Figure 4.3(r): Extraction of assembly into drawing sheet.....	17
Figure 5.2(a): Angular calibration from shifting pipes from step 1 to step 2.....	19
Figure 5.2(b): Angular calibration from shifting pipes from step 2 to step 3.....	19
Figure 5.3: Inclination of pipes with vertical.....	21

List of acronyms and abbreviations

PVC:	Polyvinyl Chloride
CVC:	Chlorinated Polyvinyl Chloride
MOALD:	Ministry of Agriculture and Livestock Development
NCFD:	National Centre for Fruit Development
UNECE:	United Nations Economic Commission for Europe
FAO:	Food and Agriculture Organization of the United Nations
GB/T:	Guóbiāo Tuījiàn (Term used for National Standard in China)
AGMARK:	Agriculture Mark
DMI:	Directorate of Marketing and Inspection
mm:	Millimetre
NRs:	Nepali Rupees
Kg:	Kilogram
Sec:	Second
Min:	Minute

CHAPTER ONE: INTRODUCTION

1.1 Background of Project

Since the childhood, we have reading and saying, Nepal is an agriculture country. Around 62% of the total population are involved in agriculture. Fruit farming is one of the backbone of agriculture in Nepal .Though Nepal has immense potential in fruit farming, farmers in Nepal faces challenges in post-harvest loss, quality, and market access. One of the major challenge in this sector, sorting mechanism of fruits. There is no proper guidelines and rules about fruit sorting in Nepal. In most of the places, the sorting is done manually by visual inspection. The fruit sorting machine are very expensive as shown in figure below. They are not affordable to small villagers and local vendors. Seeing this grass root problems and cost of fruit sorting machine, the idea of project was developed to build an adjustable user-friendly and low cost fruit sorting machine.

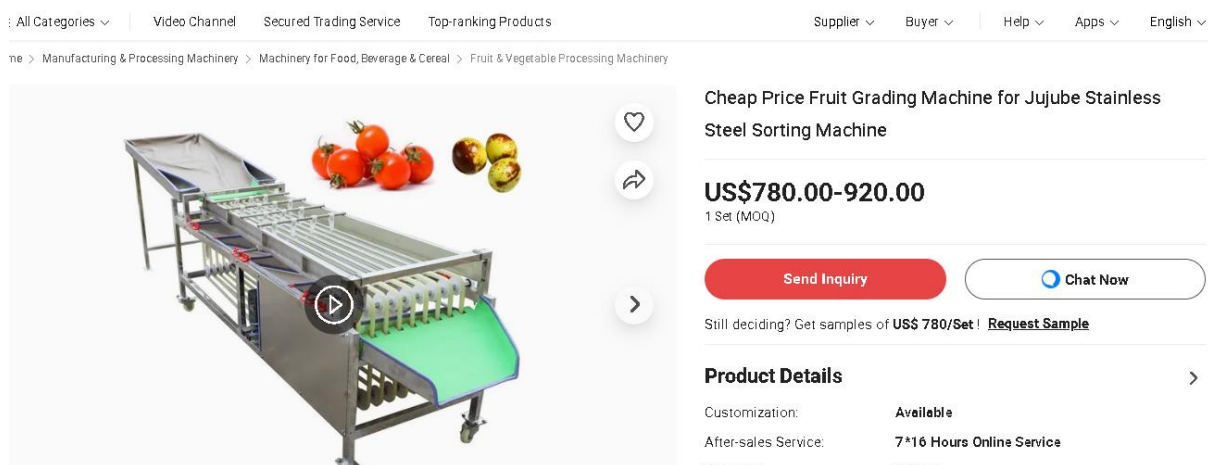


Figure 1.1(a): Fruit grading machine with price (Alibaba Machinery Categories, 2024)

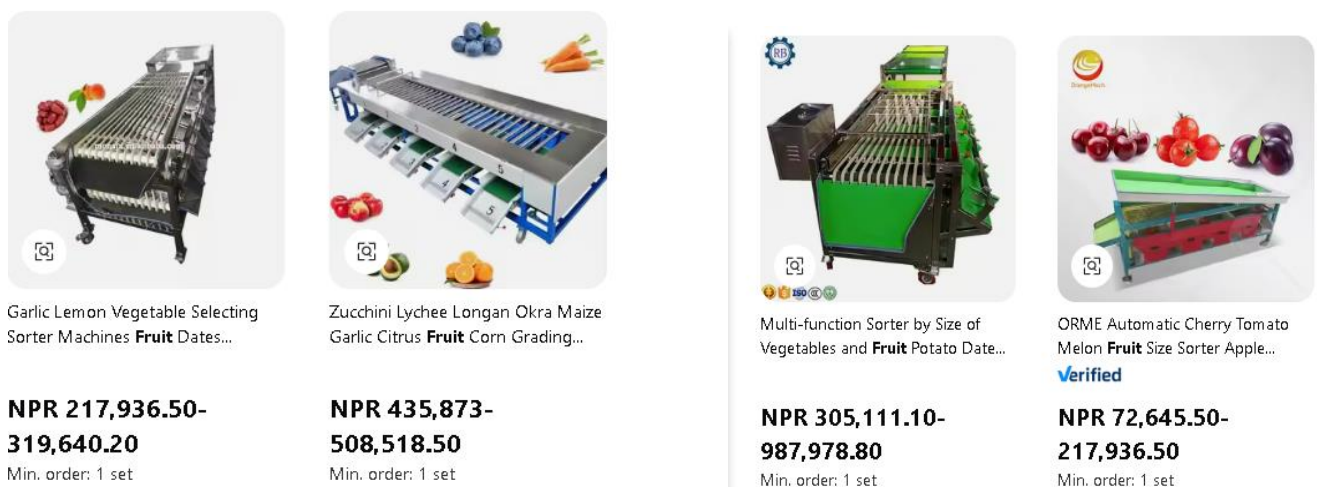


Figure 1.2(b): Fruit grading machine with price (IndiaMART Agriculture Machinery, 2025)

Figure 1.1(c): Fruit sorting machine with its price (IndiaMART Agriculture Machinery, 2025)

Statistics of orange production in economic year 2076/77

Province	Total area (hector)	Productive area (hector)	Production (ton)	Productivity hector/ton
Koshi	5996	3603	39639	10.93
Madhesh				
Bagmati	4348	2478	23130	9.34
Gandaki	7795	3534	39226	11.10
Lumbini	3089	1791	20507	11.45
karnali	3617	2118	23240	10.97
SudurPashhim	1746	1027	10708	10.43
Total	26591	14551	156180	10.73

Table 1.1: Statistics of orange production in economic year 2076/77 (NCFD, 2076/77)

Form this table, we can clearly observe to potential of orange faming in Nepal. The average orange production in Nepal 156180 tons with productivity 10.73 which is very low. Not only this, the post production facilities are very low. There is no proper storage center, no proper market and sorting mechanism. There is no proper grading guidelines of those oranges. They are sorted in size by people in the market. The middlemen buy fruits from farmers and supply to market. Then, in the markets they are separated based on size like big, small, and medium by people. The case is almost similar for most of the fruits produced in Nepal. So, we aim to make a very small change by designing simple but effective sorting machine.

Fruit sorting machines are crucial for grading the fruits. It is very useful to farmers, vendors, whole seller, and retailers as well. When we go to fruit shops, we pick fruit based on its size. For instance, when we buy tomatoes, we buy neither too small tomatoes nor too big. The same thing happens when we buy apples, oranges, lemons, etc. So, it is clear that sorting is necessary so that customers will get an easy experience while buying fruits. This was a driving factor to pursue this project.

As we know, fruits are one of the most valuable and delicate agricultural products. Their quality, appearance, and uniformity in size play a significant role in determining market value and consumer preference. In countries like Nepal, where agriculture is a major source of livelihood, fruit farming contributes significantly to household income and food security.

However, one of the persistent challenges faced by farmers and small-scale vendors is the lack of access to efficient and affordable post-harvest handling equipment, especially for size-based sorting.

This situation highlights the need for an alternative solution, a low-cost, user-friendly, and power-free mechanical fruit sorting machine that can effectively separate fruits based on size. A gravity-driven system using adjustable shafts and protective cushioning offers a practical and economical approach. Not only would such a machine help reduce labour and post-harvest losses, but it would also improve the consistency of sorted fruits, increase market competitiveness, and benefit small-scale producers and vendors who operate on tight margins. This project aims to design and develop such a mechanical sorting machine that can be widely adopted in the agricultural sector, particularly in regions where resources are limited but the need for improved post-harvest handling is high. It is mostly used in wholesale fruits and vegetables shop like Kalimati, Balkhu etc., and different farmers group and committee where the fruits are sold collectively, where tons of vegetables and fruits are sold every day. So in such a place, fruits and vegetables need to be sorted based on size to grade them and make sure all of them are sold by changing price based on grade.

1.2 Statement of Problem

Wholesale fruit and vegetable markets like Kalimati and Balkhu handle a large amount of produce every day. These places are important because fruits and vegetables are collected there before they are sent across the Valley. Even though the volume is high, most of the sorting work is still done by hand. Workers have to separate fruits based on size, and this creates many problems. During busy hours, manual sorting becomes slow and causes long delays. It also increases labor cost, which affects the profit of wholesalers who already work with low margins. When large amounts of fruits arrive at once, workers sometimes rush the job, and this can damage fruits, which reduces their quality and leads to losses.

Traditionally, fruits are sorted manually based on size and appearance. This process is not only time-consuming and inconsistent but also physically demanding. Moreover, manual sorting increases the chances of mishandling, leading to bruising or other damage that can reduce the quality and shelf life of the fruits. These issues are particularly critical when supplying to wholesale markets, exporters, or organized retailers where uniformity and presentation are important. Although modern sorting machines with advanced sensors and automated systems are available in developed countries, they are usually expensive, require electrical power, and are designed for large-scale operations. Such machines are beyond the financial and technical

reach of smallholder farmers and local cooperatives, especially in rural areas with limited infrastructure.

At present there are many farmers group in our country where they produced different types of fruits and sell together through a committee or cooperative. They sell their products collectively and set the prices for the benefits of farmers. In such scenario, a simple fruit sorting machine will be very useful. It will help farmers to grade their fruits properly set price based on the size for each category. So the machine is very useful for different farmers groups across rural area. There are modern sorting machines that can do this job, but they are very expensive and need a lot of space and electricity. These machines are usually made for big industries, not for individual wholesale shops or medium-size market operators. This means, many local sellers cannot afford them. Because of this, there is a clear need for a simple and affordable mechanical sorting machine that does not require power. Such a machine should be easy to adjust, so it can handle different fruit sizes quickly and consistently. If wholesalers had access to this kind of machine, they could work faster, and improve the way fruits are handled and sold.

1.3 Objectives

1.3.1 General Objective

To design, fabricate, and validate a highly adjustable, user-friendly, and cost-effective mechanical fruit sorting machine.

1.3.2 Specific Objectives

- To design and develop an accessible sorting mechanism
- To execute practical engineering and fabrication
- To validate performance and reliability
- To compare manual sorting and machine sorting.

1.4 Scope of Sorting Machine

- **Useful for local farmers and vendors:** This machine can be used by small farmers, local vendors, and wholesalers who cannot afford expensive automated machines.
- **Reduction of post-harvest losses:** By sorting fruits smoothly and carefully, the machine can reduce bruising and damage. This helps maintain fruit quality and reduces losses after harvest.
- **Improved grading and fair pricing:** Sorting fruits by size helps in proper grading. Farmers and sellers can set different prices based on grades, which can increase overall income. In this way, the most of the produced fruits can be sold which is very beneficial to farmers. The different grades of fruits will be a choice for customer as well.

- **Low cost and easy fabrication:** The machine is made using simple materials and basic workshop tools. This makes it easy to repair, modify, or reproduce in local workshops. The design and fabrication is not very complex even the lay man can easily understand and fabricate the machine which very help to commercialization.
- **Future improvement and modification:** The design can be further improved by adding better adjustment mechanisms, increasing capacity, or partially automating some functions in the future. The modification of the machine will make further useful and solve real grass root problems faced by farmers, local vendor, wholesalers as well as retailers.

1.5 Key Features

- **Power Free Operation:** The machine operates entirely using gravity, requiring no electricity or fuel. This makes it ideal for use in rural or off-grid farming areas.
- **Size Based Sorting:** Using Adjustable Pipes: Fruits are sorted by diameter using a set of pipes with adjustable and non-uniform gaps. This allows sorting into different size categories with improved precision.
- **Customizable for Different Fruits:** The gap between the shafts can be adjusted manually, making the machine suitable for sorting various types of fruits such as apples, oranges, guavas, lemons, and more. The inclination of pipes can be adjusted too so the flow rate can be controlled.
- **Simple and Low Cost Design:** The structure is made using commonly available materials and basic mechanical components, making the machine affordable and easy to manufacture locally.
- **Manual Feeding with Continuous Flow:** Fruits are placed into a hopper or bin at the top, from where they roll down naturally over the shafts, allowing a steady and continuous sorting process.
- **Portable and Compact Structure:** The machine is designed to be lightweight and easy to move, making it suitable for small farms, collection centers, or mobile use.
- **Easy Maintenance:** With no complex electronics or moving parts requiring lubrication or calibration, the machine is easy to clean, repair, and maintain.
- **Labour Saving:** Reduces the need for manual sorting labor, thereby saving time and effort while maintaining consistency in output.
- **Environmentally Friendly:** Since it does not use any form of external energy, the machine is eco-friendly and contributes to sustainable farming practices.

CHAPTER TWO:

LITERATURE REVIEW

Fruit sorting and grading is a key post-harvest operation that directly affects fruit quality, market price, and shelf life. Proper grading helps farmers and vendors separate fruits into different size categories so that pricing becomes fair and uniform. In many developing countries, including Nepal, fruit sorting is still carried out manually. We visited different official website like National Centre for Fruit Development (Publications, 2023) ,and Ministry of Agriculture and Livestock Development etc., and we find out there is no proper rules and guidelines for fruit sorting and grading in Nepal. (Information Centre, 2025)

Many countries and international organizations have established clear guidelines for grading and sorting spherical fruits such as apples, oranges, and lemons. These guidelines mainly focus on size, shape, color, and allowable defects, and they are used to maintain uniform quality in markets. In countries like the United States, grading standards defined by the USDA classify fruits into grades such as U.S. Fancy, U.S. No. 1, and U.S. No. 2, with size usually measured by equatorial diameter or pack count (Trade and Market, 2018). Similar systems exist in the European Union, where mandatory marketing standards specify minimum diameters, size codes, and uniformity limits for apples and citrus fruits. International organizations such as Codex Alimentarius and UNECE also provide widely accepted standards that many countries follow for domestic trade and exports. (Publication, 2022)

Asian countries like India and China have also developed national grading systems. India follows AGMARK standards, which define grades like Extra Class, Class I, and Class II based on fruit size, shape, and defects. (Publications, 2021) China uses national GB/T standards where fruit size is classified mainly by diameter, along with quality and appearance requirements. Across all these systems, size-based sorting plays a major role, and mechanical sorting methods such as roller or shaft-based graders are commonly used to achieve uniformity. Several studies show that most mechanical fruit sorting machines operate based on physical and mechanical properties of fruits such as diameter, sphericity, mass, surface friction, and rolling behavior. (Elkaoud & Mahmoud, 2022) designed a simple mechanical fruit sorting machine using adjustable rods to classify apples, oranges, and plums into size categories. Their experimental results showed that sorting accuracy depends strongly on fruit shape. Plums, which are nearly spherical, achieved sorting efficiencies up to 98%, while apples showed lower

accuracy due to irregular shape. This study proves that a simple adjustable rod-based mechanism can achieve high accuracy without the need for sensors or electronic components. (Ahmadi, Bazyar, Cavallo, & Ziyadeh, 2020) developed a belt-and-roller based fruit grading machine for apples and oranges. In their design, fruits move on a belt and fall through roller gaps of increasing size. Their results showed sorting accuracy above 90% under optimized operating conditions. The study also reported that feeding speed has a major effect on performance. Lower belt speed improved sorting accuracy and reduced fruit damage, while higher speed caused bruising, especially in soft fruits. These findings highlight the importance of controlling speed and spacing, which is directly relevant to gravity-driven mechanical sorting systems like the one proposed in this project.

Roller-based grading systems have also been studied extensively. (Fager, 1918) developed a rotating roller grading machine for citrus fruits such as Kinnow and Mosambi. The machine achieved grading efficiencies above 80% at optimal roller speed. The study showed that increasing speed beyond a certain limit reduces accuracy because fruits do not have enough time to fall through the correct gaps. This observation supports the idea that simple mechanical machines must balance speed and accuracy, which is a key design consideration in adjustable shaft-based sorting machines.

Some researchers focused on gravity-based and inclined sorting mechanisms. (Elkaoud & Elglaly, 2019) designed a low-power grading machine for Valencia oranges using inclined surfaces and frictional differences. Their work emphasized that fruit properties such as coefficient of friction, projected area, and volume must be considered during design. Although image-processing-based grading systems were found to be more accurate, the study concluded that mechanical systems are more practical for small farmers due to lower cost and simpler operation. This supports the choice of a non-electric, mechanical sorting system in the present project.

Other studies addressed the issue of fruit damage during mechanical sorting. (Khalil, Atallah, & Ebaid, 2012) developed a drum-type grading machine for apricots and found that rigid materials like PVC caused bruising. When rubber coating was applied, fruit damage was significantly reduced, and grading efficiency reached more than 98%. This study highlights the importance of cushioning and gentle handling, which is a limitation acknowledged in the current prototype and can be improved in future versions.

(Awasthi, 2021) developed a low-cost adjustable grading machine using conveyor belts and rotating bars for fruits and vegetables. The design focused on adjustability, ease of fabrication,

and reduction of fruit damage. The study emphasized that machines intended for small farmers should be flexible and capable of handling different crops. This concept aligns well with the objective of the present project, which aims to design an adjustable machine suitable for different fruit sizes.

Review studies such as that by (Joshi & Awate, 2016) compared mechanical grading systems with image-based grading techniques. While image processing systems offer higher accuracy and can detect defects and color variations, they are expensive, require controlled lighting, and need skilled operation. Mechanical graders, on the other hand, are easier to maintain, consume less energy, and are more suitable for rural and resource-limited areas. The review concluded that mechanical grading remains the most appropriate solution for small-scale farmers and local markets.

From the reviewed literature, it is clear that mechanical fruit sorting machines can provide reliable performance when designed properly. Most studies agree that sorting accuracy depends on fruit shape, shaft or roller spacing, operating speed, and surface material. Although advanced automated machines exist, they are not accessible to small farmers and local wholesalers due to high cost and power requirements.

Therefore, there is a clear gap for a simple, low-cost, adjustable, and non-electric fruit sorting machine. The present project aims to address this gap by developing a gravity-driven adjustable shaft-based sorting machine suitable for local markets and small-scale agricultural use.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Theoretical Framework

This project started by looking at real problems seen in fruit sorting, especially the issues faced by farmers and local vendors. We wanted to work on something that is useful in real life, not too expensive, and possible to make with limited resources. The main idea was to choose a project that could actually help at the ground level.

At the beginning, we discussed many project ideas within the group. We used simple methods like brainstorming and group discussion to share thoughts and suggestions. Reverse brainstorming was used to understand what is wrong with existing systems. We also made mind maps to connect ideas and see which ones made sense. Tools like SWOT analysis and the SCAMPER method helped us check whether an idea was practical, affordable, and easy to build. After comparing different ideas, we decided to work on an adjustable fruit sorting machine. This option felt more realistic because it solves a common problem and does not depend on electricity. In the early stage, we first drew rough sketches on paper to understand the shape, layout, and working of the machine. Along with this, we did background study and reviewed related literature, which helped us move forward to detailed design, fabrication, and data collection.

3.2 Data Collection

3.2.1 Primary data Collection

Primary data were collected by directly going to the field. We visited wholesale markets like Kalimati and Balkhu to see how fruits are actually sorted in daily practice. We also visited some retail fruit shops and talked with farmers and vendors to understand their problems, labor needs, and time required for sorting. Along with this, we checked the prices of materials such as angles, metal sheets, and rods from local shops to estimate project cost. (Category, 2025) We also tried sorting fruits manually and then using our machine, and noted the time difference to get a basic idea of efficiency. (Market Information, 2025)

3.2.2 Secondary Data Collection

Secondary data were collected by reading existing materials related to fruit sorting. We studied journal papers, previous project reports, literature reviews, and some news articles to understand how fruit sorting machines work. We also read about grading and sorting guidelines followed in other countries and the performance of machines developed by other researchers. This helped us understand standard practices and improve our project design.

3.3 Study Design

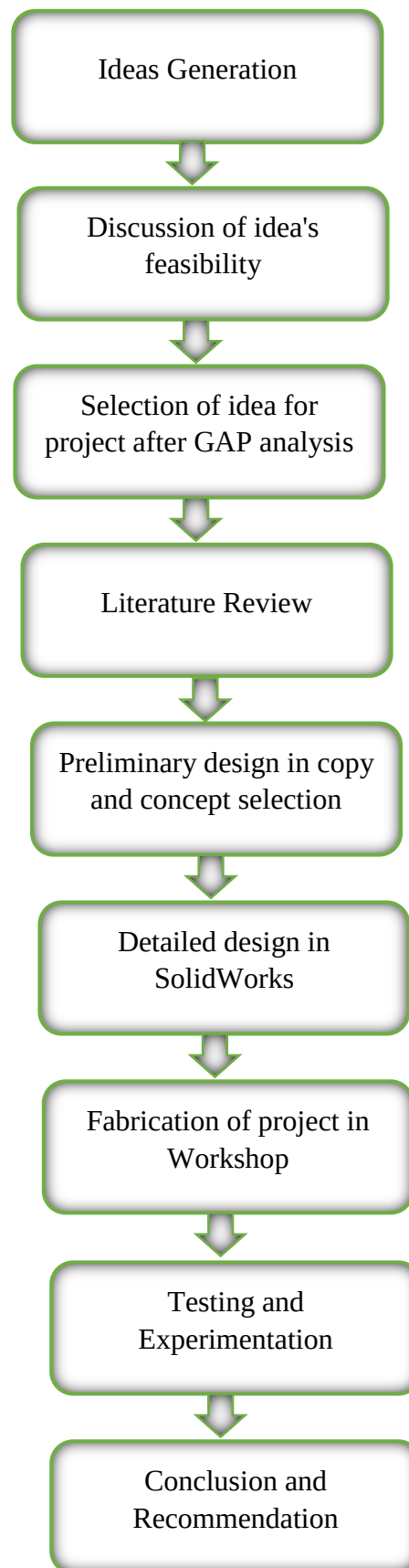


Fig 3.1: Flowchart for study design of the project

Explanation of flowchart

The project followed a structured workflow to ensure a practical, feasible, and technically sound outcome. It began with idea generation, producing concepts such as an automatic pipe cutting machine, automatic pet feeder, French fries making machine, and electric scooter. Techniques like brainstorming, mind mapping, reverse engineering, SCAMPER, and forced relationships were used.

Feasibility discussions were held through group sessions, evaluating cost, material availability, technical complexity, and suitability for local conditions. After detailed SWOT analysis, the project “Design, Fabrication, and Testing of an Adjustable Fruit Sorting Machine” was selected. This project addresses fundamental problems faced by farmers, vendors, and wholesalers, involves extensive fabrication processes, and has commercial potential.

A gap analysis of existing fruit sorting methods revealed a lack of affordable, power-free machines for small farmers. The selected project addressed this real-world problem and filled a technological gap. A literature review followed, studying previous research, sorting mechanisms, grading standards, and design approaches. This helped understand limitations, benchmarks, and areas for improvement. Different government websites were visited to find standard rules for fruit sorting in different countries.

In the preliminary design stage, rough sketches visualized the structure, working principle, and layout. Multiple design options were explored, and suitable dimensions and mechanisms were selected. The concept was then developed in SolidWorks, detailing dimensions, component arrangements, and assembly, allowing early detection of design issues.

Fabrication was carried out using locally available materials in campus workshop and standard manufacturing processes like welding, grinding, riveting, drilling etc. and, assembling components according to design specifications. After that the machine was calibrated i.e. linear calibration and angular calibration. Similarly, the inclination of pipes with vertical was also measured and calibrated. After that testing of machine was done, different fruits of different sizes were sorted manually and using machine at different inclination angle and corresponding time to sort the fruits were noted respectively.

Finally, conclusions and recommendations were drawn based on experimental findings. The machine’s performance was assessed against project objectives, limitations were noted, and suggestions for future improvements were provided, completing the project development cycle systematically. Overall, all the steps mentioned in above flow chart was carried out to meet the objectives of the project.

CHAPTER FOUR:

DESIGN AND FABRICATION

4.1 Preliminary Design

At first the preliminary design of machine was done in paper. It helps to visualize the machine and helps to generate idea. It was foundation of next phase of design in SolidWorks. After generating concept of how the machine will work we tried to draw preliminary sketches in paper and take it as a reference for future designs.

4.2 Design in SolidWorks

After preliminary sketches the detailed 3D design were made in SolidWorks. At first, each components were designed then assembly was done to make complete design. The preliminary designs we made at first was slightly change after.

4.3 Components Design and Fabrication

i. Frame

At first, the foundation of machine, i.e. frame was designed. At first, the frame was designed using square tube. But later the frame was made using angle of mild steel of 25×25 mm and 2 mm thickness because it was available in workshop.

Dimension of frame: length=1475 mm, breath=978 mm and height= 1000.84 mm



Figure 4.3(a): Design of frame before



Figure 4.3(b): Design of frame after



Figure 4.3(c): Actual frame fabricated in workshop

ii. Hopper

Then the next part was hopper. The hopper acts as the primary intake for sorting machine. The design of hopper was also changed later. Rings slot was added to hopper later so the fruits fell down from hopper to pipes without any damage. The rings acts as resting slot for the pipes. The hopper was made by using flat bar frame of 2 mm thickness. Then the main part of hopper was using metal sheet and it was placed on frame (cover) of hopper. The hopper attached to frame and base was changed later.

Dimension of hopper frame which is in the shape of triangular base prism.

First side (a) = 610.01mm Second side (b) = 338mm Third side (c) = 677.59 mm

Length or height (l) = 640mm True radius of ring(R) = 14 mm



Figure 4.3(d): Design of hopper frame before



Figure 4.3(e): Design of hopper frame after



Figure 4.3(f): Design of hopper cover



Figure 4.3(g): Actual hopper frame fabricated in workshop

iii. Sliding Sheet

The next components of the project is sliding sheet. It was fabricated using metal sheet. It is the part of machine where the fruits fall down after sorted through pipes and the fruits roll down to the collection bins. It is inclined so the fruits can be rolled down easily. Square flat tubes were used for its support.

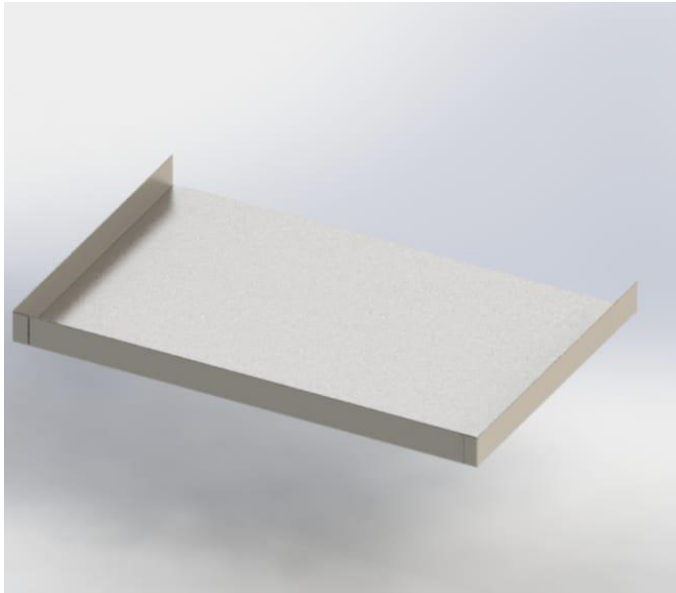


Figure 4.3(h): Design of Sliding Sheet

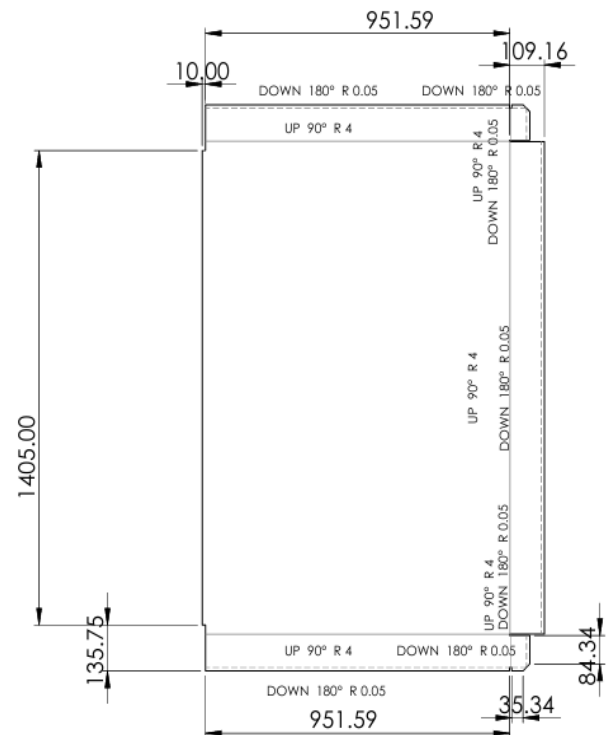


Figure 4.3(i): Detailed design of sliding sheet

iv. Slider for Pipes

This is one of the major components of the machine. This part is what it makes possible for adjustable fruit sorting machine. So metal flat sieve (plate) was taken then a gap was cut using grinder. In that gap the pipes are placed using nut, bolt, and washer. So the pipes can be easily slide in that gap making it possible to adjust gap between two pipes.



Figure 4.3(j): Design of slider for pipes

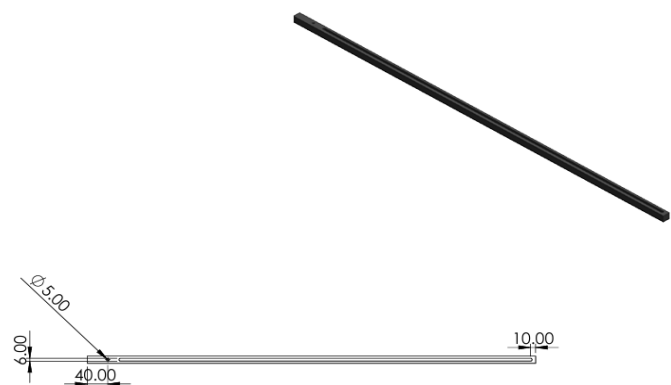


Figure 4.3(k): Detailed design of slider for pipes

v. Pipes

The pipes was essential part of machine. We did research on what type of pipes should be used. At first, we decided to use PVC pipes, but it was not strong. Then we checked out CPVC pipes and it was expensive and it comes in roll of 3 metre. But we need at least 1.6 m for each piece so it was discarded considering the resources wasted will be maximum. Then at last, we used aluminum pipes because it was light weight but strong enough to hold load for our prototype. and it was within the budget. Dimension of pipe are 1700m with internal diameter 22.5mm.

vi. Nut, Bolt and Washer

Different sizes of nuts and bolts were used for each purposed. We used M5, M10 and M12 nut and bolts for fastening the components. Washer were used for protection and make sure the components are tightly fastened.

vii. Square Rod

Square rods were used to make the frame stable. They were welded at the bottom of frame to make the structure stable. Dimension is 1765 mm of 10×10 mm



Figure 4.3(l): Aluminum pipes used for sorting



Figure 4.3(m): M5 nut, bolt, and washer



Figure 4.3(n): Square rod used to make frame stable

4.4 Assembly of each components

After designing of each components in SolidWorks, they were assembled together using assembly tool is SolidWorks. It was very difficult job and when the assembling was done it gives clear idea of the machine. Then the machine was fabricated in workshop based on that 3D assembled sketch made in SolidWorks.



Figure 4.3(o): Assembly of all components in SolidWorks



Figure 4.3(p): Actual machine fabricated in workshop

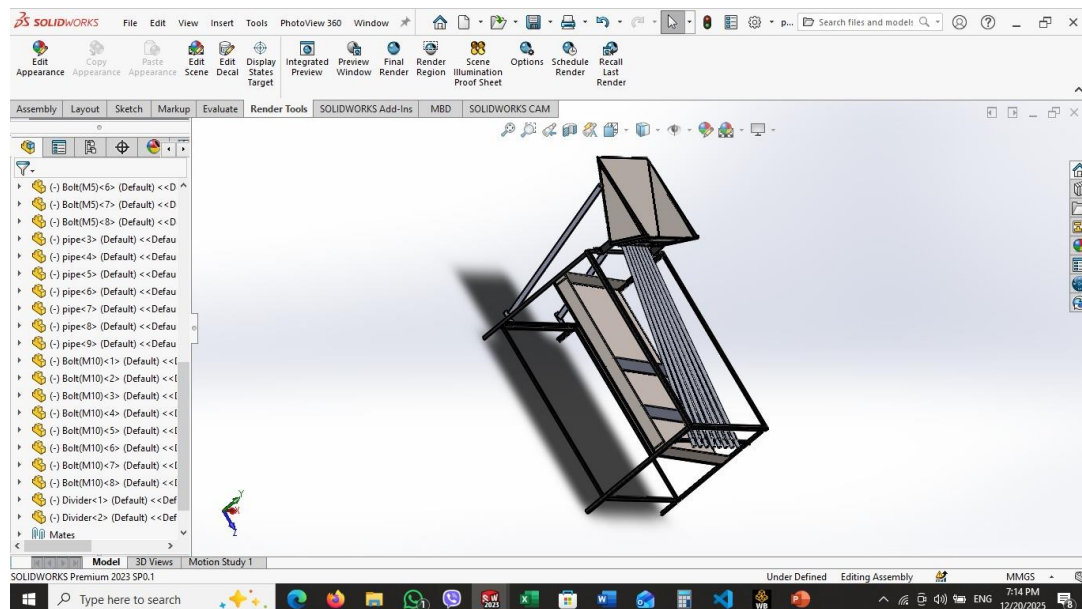


Figure 4.3(q): Rotating view of assembly in SolidWorks

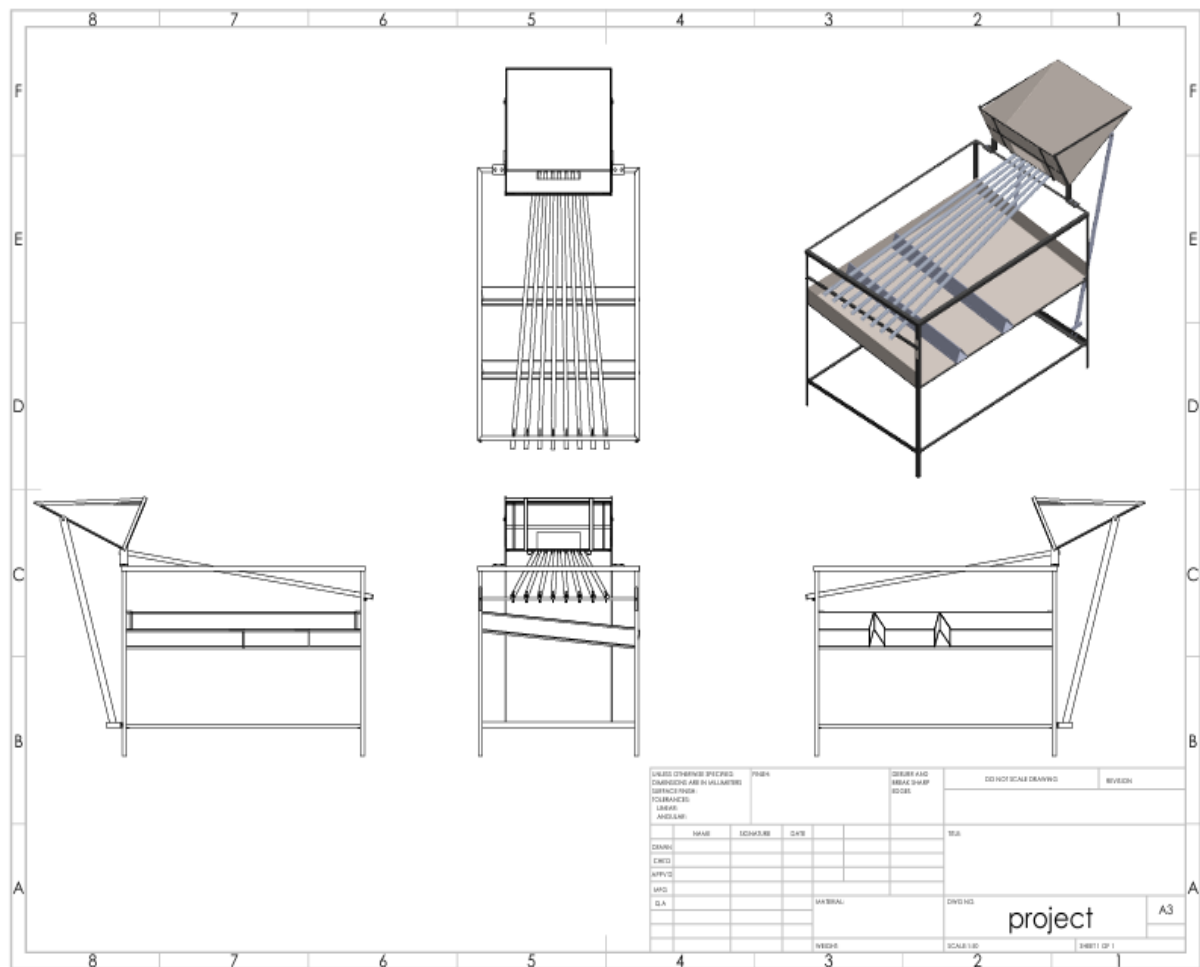


Figure 4.3(r): Extraction of assembly in drawing sheet

4.5 Machine Used in Fabrication

- Welding machine
- Drilling Machine
- Hand driller
- Grinding machine
- Sheet Bending machine
- Riveting Tools
- Measuring Tools: Vernier caliper, Measuring tape, Steel scale etc.
- Hand Tools: Hammer, Spanner, Pliers, files etc.

CHAPTER FIVE:

CALIBRATION AND TESTING

5.1 Calibration of Machine

After successfully fabricating the machine, the next step was its calibration. The calibration of machine is very important so it can be used by lay man easily. We establish a reliable relationship between angular position of adjustable pipes and corresponding gap size between pipes. For calibration the gap between two pipes was divided into three steps manually. Each step represent different gap setting. Those three steps were marked permanently. The three steps are:

Step 1: The gap between two sliding pipes in slider is maximum

Step 2: The gap between two sliding pipes in slider is medium

Step 3: The gap between two sliding pipes in slider is minimum

This steps are set by us manually based on some research of fruit size like apples, oranges, tomatoes, lemons, etc. This steps can be further divided, but in our machine we set only 3 steps. After marking those steps we categorized three compartment large, medium and small. Then when the pipes were at step 1 the minimum and maximum gap in each compartments were measured. We conducted the same procedure for step 2 and step 3 and corresponding gap at each compartments were noted which is shown in below table.

S.N	Compartments					
Steps	Small(mm)		Medium(mm)		Large(mm)	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
1	14	57	58	77	78	92
2	14	45	46	63	64	75
3	14	33	34	41	42	49

Table 5.2: Calibration table for adjusting the pipes

From the table we observed that when the two pipes were set at step 1, in small compartment the minimum and maximum gap were 14 mm and 57 mm. In medium compartment the range of fruits that can be sorted is 58mm to 75mm. In large compartment the range is 78mm to 92mm. So the fruits whose size varies in above manner can be sorted by setting machine at step 1. From the table, it can be observed that when the two pipes were set at step 2, in the small compartment the minimum and maximum size of fruits that can be sorted were 14 mm and 45 mm. In the medium compartment, range is 46 mm to 63 mm. Similarly, in the large

compartment, fruits having sizes between 64 mm and 75 mm can be sorted. Therefore, fruits falling within these size ranges can be sorted by adjusting the machine to step 2.

Likewise, when the pipes were adjusted to step 3, the small compartment fruits with sizes ranging from 14 mm to 33 mm can be sorted. In the medium compartment, the range was 34 mm to 41 mm. For the large compartment, fruits with sizes between 42 mm and 49 mm can be sorted. This shows that setting the machine at step 3 is suitable for sorting relatively smaller fruits into three size categories.

5.2 Angle calibration

In the machine only the linear calibration is done but it is necessary to calculate how much angle change when the gap between two pips is changed.

Calculation

Length of pipe = 1700 mm

Gap Changes

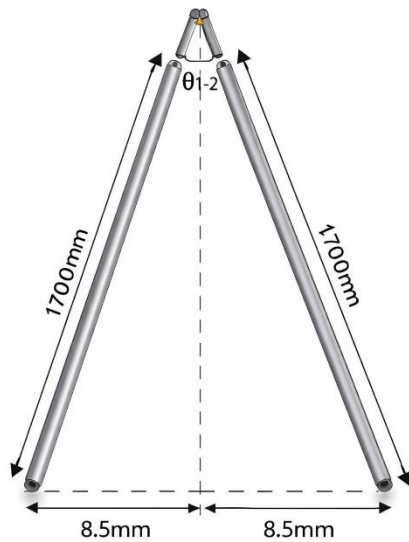
Step 1-2

$$h_{1-2} = 92-75 = 17\text{mm}$$

Step 2-3

$$h_{2-3} = 75-49 = 26\text{ mm}$$

Changing pipes
from step 1 to step 2



Changing pipes
from step 2 to step 3

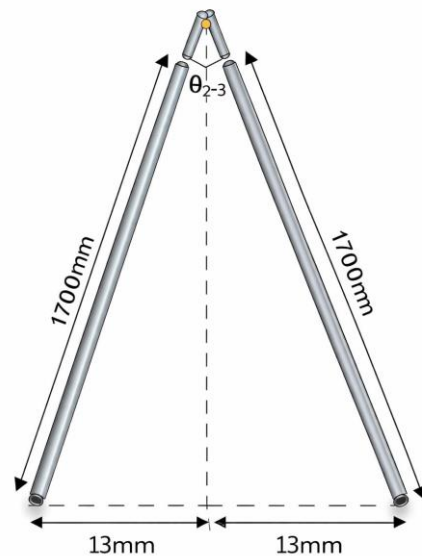


Figure 5.2(a): Angular calibration from shifting pipes from step 1 to step 2

Figure 5.2(b): Angular calibration from shifting pipes from step 2 to step 3

Angle shift calculation

$$\sin\theta_{1-2} = 2 \times \frac{8.5}{1700} = 0.01^\circ \quad \sin\theta_{2-3} = 2 \times \frac{13}{1700} = 0.015^\circ$$

$$\theta_{1-2} = \sin^{-1}(0.01) = 0.573^\circ \quad \theta_{2-3} = \sin^{-1}(0.153) = 0.859^\circ$$

Angle Change per mm

Step 1-2

Step 2-3

$$0.573^\circ/17 = 0.034^\circ/\text{mm}$$

$$0.859^\circ/26 = 0.034^\circ/\text{mm}$$

Gap Change (mm)	Angle change (degree)
1	0.034
10	0.34
50	1.70
100	3.40
200	6.80
300	10.20

Table 5.2: Angle calibration table per mm

5.3 Inclination (Slope) Adjustment mechanism of pipes

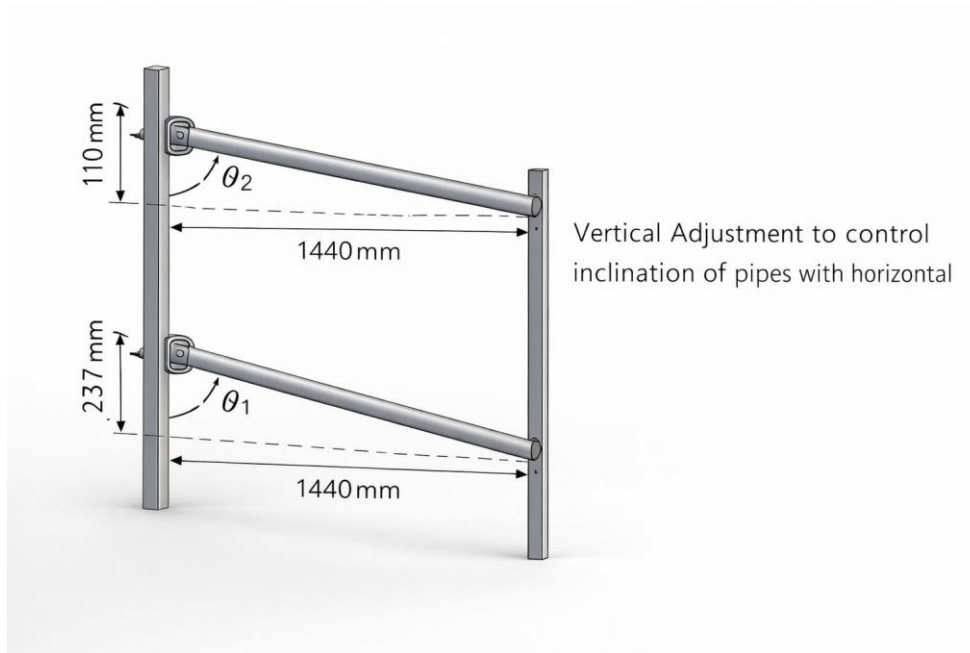


Figure 5.3: Inclination of pipes with vertical

For maximum inclination with vertical

Vertical length (p) = 1440mm

Distance correspond to vertical sliding slot (b) = 237 mm

$$\tan\theta_1 = \frac{1440}{237}$$

$$\theta_1 = \tan^{-1}\left(\frac{1440}{237}\right) = 80.65^\circ \approx 81^\circ$$

For minimum inclination with vertical

Vertical length (p) = 1440mm

Distance correspond to vertical sliding slot (b) = 110mm

$$\tan\theta_2 = \frac{1440}{110}$$

$$\theta_2 = \tan^{-1}\left(\frac{1440}{110}\right) = 85.43^\circ \approx 85^\circ$$

The maximum and minimum inclination of pipes with vertical found to be 81° and 85° . This inclination can be adjusted between these angles (81° - 85°) by moving the pipes up and down in vertical slider to control the flow rate of rolling fruits.

5.4 Testing of machine

To evaluate the machine, 5 kg of oranges, 6 kg of apples, and 6 kg of guavas of varying sizes were used. The fruits were first sorted manually by multiple students using visual inspection, and the sorting time was recorded. The same fruits were then sorted using the machine by keeping pipes at angles of 85° and 81° with vertical to observe flow rate when the inclination of pipes changed. Sorting was repeated 15 times at each angle, and the time taken was noted. The data was then filled in observation table.

5.4.1 Manual sorting

Iteration (By different students)	Time taken to sort 5 kg oranges (min)	Time taken to sort 6 kg apples (min)	Time taken to sort 6 kg guavas (min)
1	1.46	1.48	1.4
2	1.41	1.55	1.35
3	1.45	2.1	1.41
4	1.25	1.38	1.3
5	1.38	1.45	1.32
6	2	1.3	1.25
7	1.32	2.05	1.29
8	1.4	1.55	1.38
9	1.45	1.56	1.42
10	1.38	1.39	1.45
11	1.4	1.47	1.3
12	1.35	1.5	1.29
13	1.41	1.38	1.3
14	1.42	1.52	1.3
15	1.15	1.5	1.43
16	1.28	1.42	1.29
17	1.32	1.34	1.3
18	1.43	1.58	1.35
19	1.56	2.1	1.5
20	1.45	1.28	1.42
21	1.33	1.48	1.27
22	1.19	1.35	1.3
23	1.27	1.4	1.36
24	1.34	1.37	1.43
25	1.49	2.15	1.39
26	1.5	1.41	1.34
27	1.47	1.53	1.34
28	1.43	2	1.39
29	1.39	1.44	1.25
30	1.46	1.35	1.3
Average	1.404	1.546	1.347

Table 5.3.1: Manual sorting time of different fruits by different people

5.4.2 Sorting by machine

The same 5 kg oranges, 6 kg apples, and 6 kg guavas were sorted using machine we fabricated at two different condition i.e. When the angle made by pipes with vertical was 85 degree and 81 degree respectively.

Iteration	Sorting of 5 kg oranges when inclination of pipes with vertical is (sec)		Sorting of 6kg guavas when inclination of pipes with vertical is (sec)		Sorting of 6 kg apples when inclination of pipes with vertical is (sec)	
	85 ⁰	81 ⁰	85 ⁰	81 ⁰	85 ⁰	81 ⁰
1	17.88	14.3	14.9	8.92	17.55	12.05
2	19.35	14.55	15.22	8.95	17.35	11.55
3	18.5	15.02	15.85	9.22	17.5	11.9
4	17.45	14.84	15	9.3	16.9	11.85
5	18.28	13.78	15.47	9.43	16.65	12.1
6	18.8	14.1	15.5	9.73	17.22	12.34
7	17.57	13.8	15.32	8.9	17.38	11.58
8	17.72	14.25	14.95	9.5	17.5	11.9
9	18.4	14.4	15.18	8.8	16.88	12.05
10	17.48	14.35	15.34	9.28	16.76	12.14
11	17.21	13.9	15.25	9.03	17.68	11.87
12	17.67	14.23	14.44	9.63	17.45	11.75
13	18.02	13.8	15.58	8.79	16.9	12.34
14	17.92	14.5	15.65	9.19	17.62	11.9
15	18.3	13.75	15.5	9.11	16.65	12.1
Average	18.036	14.238	15.276	9.185	17.199	11.961

Table 5.3.2: Sorting time taken by machine for different fruits at different angle

The table presents the sorting time obtained using the fabricated machine at two pipe inclination angles, 85 degree and 81 degree with respect to the vertical. The experiment was conducted on 5 kg of oranges, 6 kg of guavas, and 6 kg of apples. For each fruit, sorting was repeated 15 times at both angles, and the time taken for each iteration was recorded in seconds. The results show some variation between iterations due to differences in fruit size and movement. However, for all fruits, sorting at 81 degree consistently required less time than at 85 degree. The average sorting time for oranges was 18.036 seconds at 85 degree and 14.238 seconds at 81degree. Guavas took an average of 15.276 seconds at 85 degree and 9.185 seconds at 81 degree, while apples required 17.199 seconds and 11.961 seconds at 85 degree and 81 degree respectively. Overall, the table indicates that increasing the pipe inclination improves machine performance by reducing the sorting time for all fruits.

5.5 Comparison between Manual and Machine Sorting

For 5 kg oranges

	Average time taken for sorting 5 kg oranges in seconds (sec)		
	manually	At 85° inclination	At 81° inclination
	100	18.036	14.238
Comparison factor at different inclination		$\frac{100}{18.036} = 5.54 \text{ times}$	$\frac{100}{14.238} = 7.02 \text{ times}$
Average comparison factor	$\frac{5.54 + 7.02}{2} = 6.28 \text{ times}$		

Table 5.5(a): Comparison of manual sorting and machine sorting for 5 kg oranges

For 6 kg guavas

	Average time taken for sorting 6kg guavas in seconds (sec)		
	manually	At 85° inclination	At 81° inclination
	95	15.276	9.185
Comparison factor at different inclination		$\frac{95}{15.276} = 6.21 \text{ times}$	$\frac{95}{9.185} = 10.34 \text{ times}$
Average comparison factor	$\frac{6.21 + 10.34}{2} = 8.27 \text{ times}$		

Table 5.5(b): Comparison of manual sorting and machine sorting for 5 kg guavas

For 6kg apples

	Average time taken for sorting 6 kg apples in seconds (sec)		
	manually	At 85° inclination	At 81° inclination
	115	17.199	11.961
Comparison factor at different inclination		$\frac{115}{17.199} = 6.68 \text{ times}$	$\frac{115}{11.961} = 9.61 \text{ times}$
Average comparison factor	$\frac{6.68 + 9.61}{2} = 8.14 \text{ times}$		

Table 5.5(c): Comparison of manual sorting and machine sorting for 6kg apples

CHAPTER SIX:

BUDGET OF PROJECT

The total budget to fabricate this project is analyzed below. The cost of each component with current market rate is mentioned in table. It includes labor, miscellaneous cost and contingency cost as well.

Items	Quantity	Unit	Cost(NRs)
Mild steel square rods	10 kg	85/kg	850
Mild steel angle	20kg	95/kg	1800
Sheet metal	20sqft	120	2400
Bolts and nuts	½ kg	500/kg	250
Aluminum pipes	5 piece of 3.8 m	450/piece	2250
Washer	¼ kg	350/kg	175
Mild steel flat bar	12 kg	120	1440
Labor Cost	4	1500/day	12000
Miscellaneous			1500
Contingency			2500
Total			25165

Table 6: Budget of the project

Therefore, the total budget of the project is twenty five thousand one hundred and sixty five. Nepali rupees. The miscellaneous cost and contingency cost are very important because sometimes there is unforeseen situation while fabricating the project. In such scenario, this will be very helpful. This cost can be changed with time and place. It gives overall idea of cost involved in project. This is approximate cost, not the actual cost to fabricate the machine because the cost of materials and labor always changes with time and place. So, it is recommended to consider that factor too.

CHAPTER SEVEN: RESULT AND DSSUSSION

7.1 Result Presentation

This project successfully fabricated an adjustable fruit sorting machine. The prototype was able to demonstrate sorting different size of fruits successfully. The results of the project are presented using tables and figures that show sorting time, efficiency, and size range obtained at different pipe settings. Data were collected for both manual and machine sorting of oranges, apples, and guavas Calibration results are also presented to show how fruit size varies with different pipe gaps. These results help compare the performance of the developed machine with manual sorting.

7.2 Analysis of Result

The results obtained from testing shows that the developed fruit sorting machine performs significantly better than manual sorting in terms of time and efficiency. During testing with 5 kg of oranges, the average time required for manual sorting was found to be 100 seconds, whereas the average time taken by the machine at 85^0 and 81^0 inclination with vertical were 18.036 second and 14.238 second respectively. At 85^0 inclination the machine sort s oranges 5.54 times faster than manual sorting and at 81^0 it was found 7.02 times faster. When we took average it was found the machine can sort oranges **6.28 times** quicker than manual sorting. This shows a very large reduction in sorting time when using the machine

Similarly, during testing with 6kg of apples, the average time required for manual sorting was found to be 115 seconds, whereas the average time taken by the machine at 85^0 and 81^0 inclination with vertical were 17.199 seconds and 11.961 seconds. At 85^0 inclination the machine sorts apples 6.68 times faster than manual sorting and at 81^0 it was found 9.61 times faster. When we took average it was found the machine can sort apples **8.14 times** quicker than manual sorting.

During testing with 6 kg of guavas, the average time required for manual sorting was found to be 95 seconds, whereas the average time taken by the machine at 85^0 and 81^0 inclination with vertical were 15.276 seconds and 9.185 seconds. At 85^0 inclination the machine sort s oranges 6.21 times faster than manual sorting and at 81^0 it was found 10.34 times faster. When we took average it was found the machine can sort oranges **8.27 times** quicker than manual sorting. This show when the fruits are more spherical the machine sorts faster. The inclination also changes the sorting time. If the fruits are large, the inclination must be small and for small fruits the inclination must be more.

The calibration results also helped in understanding how pipe movement affects sorting behavior. It was observed that for every 1 mm shift of the pipe, the inclination angle changed by approximately **0.034°**. Similarly, a 10 mm adjustment resulted in a 0.34° change, and 300 mm adjustment resulted in a 10.20° change in angle. This shows that small changes in pipe position have a measurable and predictable effect on the sorting mechanism. Such control is important for adjusting the machine according to different fruit sizes.

The machine operates in three different steps, depending on the gap between the pipes. **Step 1**, which has the largest gap, allows fruits with sizes ranging from **14 mm to 92 mm** to be sorted. **Step 2**, with a medium gap, sorts fruits in the range of **14 mm to 75 mm**. Similarly, when pipes were placed at **Step 3**, with the smallest gap, the machine sorted fruits between **14 mm and 49 mm**.

These results confirm that as the gap between the pipes decreases, the maximum size of fruits that can pass through also decreases. This shows that the adjustable pipe mechanism works as intended and allows flexible size-based sorting.

The pipes can be slide vertically to change the inclination angle with vertical. The inclination angle with vertical can be change from **81° to 85°**. This shows vertical adjustment is also possible in the fabricated machine.

7.3 Outcomes

The results show that the developed fruit sorting machine performs much faster than manual sorting for different types of fruits. On average, the machine sorted oranges **6.28 times**, apples **8.14 times**, and guavas **8.27 times** faster than manual methods. Sorting performance improved for more spherical fruits and at higher inclination angles, showing that fruit shape and slope significantly affect sorting speed.

The adjustable inclination mechanism, with angles varying from **85° to 81°** with vertical, effectively controlled fruit flow rate. Calibration and pipe gap adjustment allowed fruits to be sorted into different size ranges using three steps. These outcomes confirm that the machine works as intended and provides flexible, efficient, size-based sorting at the prototype level.

7.4 Precautionary Measures

During operation, fruits should be fed slowly and uniformly to avoid congestion. Pipe gaps should be adjusted carefully and locked properly before operation. Soft fruits should be handled with care to reduce bruising. The machine should be placed on a stable surface, and moving parts should be checked regularly to ensure smooth operation. The machine can't bear load beyond its capacity which should be noted carefully.

7.5 Limitations of Project

- i. **Prototype level design:** This machine is a prototype, so it is not designed to handle a very large quantity of fruits at one time. For commercial-scale use, the capacity needs to be increased.
- ii. **Limited protection against bruising:** Although cushioning materials like rubber or foam can be added to reduce bruising, they are not included in the current model. As a result, some soft fruits may still get minor damage during sorting.
- iii. **Limited number of sorting categories:** The machine currently sorts fruits into only three groups: small, medium, and large. More compartments can be added for better grading, but this was not done in the present project.
- iv. **Manual feeding required:** Fruits need to be fed manually into the hopper. This limits the speed of operation and still requires human involvement.
- v. **Jamming of fruits:** The machine works best with round or nearly round fruits. Since, fruits are not perfectly spherical some jamming of fruits was observed while testing the machine. Therefore, Irregular-shaped fruits may not roll properly and can reduce sorting accuracy.

7.5 Future improvement

- i. Sensors can be added to count the number of fruits sorted in each compartment, which would help in record keeping and pricing.
- ii. An automatic feeding system can be developed to supply fruits uniformly and reduce manual effort.
- iii. The pipe gap adjustment can be automated using a simple motor or screw mechanism for faster and more accurate setting.
- iv. The machine size and capacity can be increased so that it can handle larger quantities of fruits in wholesale markets.
- v. Soft cushioning materials can be added on contact surfaces to reduce bruising, especially for delicate fruits.
- vi. Light vibration using a simple mechanical tapper or spring can help fruits move freely without damaging them.
- vii. Overfeeding causes fruits to pile up and jam. Fruits should be fed slowly and evenly into the hopper.

CHAPTER EIGHT:

CONCLUSION AND RECOMMENDATION

8.1 Conclusion

This project successfully achieved its primary objective of designing, fabricating, and validating a low-cost, adjustable, and power-free fruit sorting machine suitable for local agricultural conditions in Nepal. The developed prototype demonstrated that an entirely mechanical, gravity-driven system can effectively sort fruits based on size with significantly improved speed and consistency compared to traditional manual methods.

Experimental results showed a significant reduction in sorting time across all tested fruits. On average, the machine sorted oranges about 6.28 times, apples 8.27 times, and guavas 8.14 times faster than manual sorting. The results also indicated that fruit shape influences performance, with more spherical fruits achieving smoother and faster sorting.

The adjustable pipe-gap and vertical adjustment mechanisms proved reliable and flexible. Calibration established a predictable relationship between pipe movement and inclination angle, allowing effective control of fruit flow rate. While shifting the pipes the angle between two pipes changes with 0.034 degree per mm. Similarly, vertical adjustment allows the inclination of pipes with vertical from 9 degree to 5 degree. The three-step gap configuration enabled sorting of fruits into small, medium, and large categories over a size range of 14 mm to 92 mm, meeting practical grading requirements.

Overall, the machine met all stated objectives and demonstrated strong potential for use by small farmers and local wholesalers. Although the prototype has limitations in capacity and automation, it provides a practical foundation for further development to reduce labor effort.

8.2 Recommendations

Based on the outcomes of this project, it is recommended that the developed fruit sorting machine be further improved and tested with larger quantities and different types of fruits. Adding features such as cushioning, automatic feeding, and counting sensors can enhance performance and reduce fruit damage. Local farmers, cooperatives, and small wholesalers can adopt this type of machine to improve sorting efficiency at low cost. With minor modifications, the machine can be scaled and customized to meet the needs of different market conditions.

Bibliography

- Khalil, K., Atallah, M. M., & Ebaid, M. T. (2012, April). DEVELOPMENT OF A GRADING MACHINE FOR APRICOT FRUITS. *Misr Journal of Agricultural Engineering*, 29(2), 510-530. doi:<https://doi.org/10.21608/mjae.2012.102005>
- Ahmadi, V. F., Bazyar, P., Cavallo, E., & Ziyaee, P. (2020). Development and Testing of a Low-Cost Belt-and-Roller Machine for Spheroid Fruit Sorting. *AgriEngineering*. doi:<https://doi.org/10.3390/agriengineering2040040>
- Awasthi, A. (2021). *Development and fabrication of a power operated low-cost small scale fruit and vegetable grading machine*. Punjab Agricultural University, Ludhiana, 141 004, India, PROCESSING AND FOOD ENGINEERING. Retrieved from <https://krishikosh.egranth.ac.in/server/api/core/bitstreams/3278b903-0ab1-4473-9eff-a0e949222c47/content>
- Categories. (2024). Retrieved from Alibaba.com: <https://www.alibaba.com/showroom/fruit-sorting-machine.html>
- Category. (2025). Retrieved from Daraz.com: <https://www.daraz.com.np/catalog/?q=aluminium%20pipes>
- Elkaoud, N. M., & Mahmoud, R. K. (2022). Design and implementation of sequential fruit size sorting machine. (R. L. Souza Ferraz , & h. R. Raj Gheyi, Eds.) *Revista Brasileira de Engenharia Agrícola e Ambiental*, 7. Retrieved from <https://www.scielo.br/j/rbeaa/a/x6LDtqjW7sGgWZGppkYvnXB/?lang=en>
- Elkaoud, N. S., & Elglaly, A. M. (2019). Development of Grading Machine for Citrus Fruits (Valencia Orange). *Journal of Soil Sciences and Agricultural Engineering*, 10(11), 671-677. doi:<https://doi.org/10.21608/jssae.2019.62864>
- Fager, G. E. (1918). *Studies on Mechanical Fruit Grading Machines*. Retrieved from <file:///S:/Project/studiesonmechani00fage.pdf>
- Fruit and Vegetable Proceesing Machine. (2025). Retrieved from IndiaMART: <https://dir.indiamart.com/impcat/fruit-grading-machine.html>
- Guiddlines. (2023). Retrieved from Chinese Standard GB/T, GBT, GB: <https://www.chinesestandard.net/>
- Information Centre. (2025). Retrieved from Ministry of Agriculture and Livestock Development: <https://moald.gov.np/>
- Joshi, A., & Awate, N. P. (2016, October). Review Paper on Physical and Mechanical Properties of Citrus Fruits and Various Techniques used in Fruit Grading System Based on their Sizes. *International Journal of Science Technology and Engineering*, 3(4). Retrieved from file:///S:/Project/Review_Paper_on_Physical_and_Mechanical.pdf
- Market Information. (2025). Retrieved from Kalimatimarket.com: <https://kalimatimarket.gov.np/>
- Publication. (2022). Retrieved from Food and Agriculture Organization Of United Nations: <https://www.fao.org/home/en>

Publications. (2021). Retrieved from Directorate of Marketing and Inspection:
<https://dmi.gov.in/GradesStandard.aspx>

Publications. (2023). Retrieved from National Fruit Development Centre:
<https://ncfd.gov.np/en>

Trade and Market. (2018). Retrieved from U>S Department Of Agriculture:
<https://www.usda.gov/>

ANNEXES

